DBMS • CORE CONCEPT NOTES

Data Warehousing and Data Mining

A complete, exam-focused study guide with diagrams, tables & revision sheet

Topics Covered

1. Concept of Data Warehousing
2. Architecture of Data Warehousing
3. Various Tools in Data Warehousing (ETL, OLAP, BI)
4. Concept of Data Mining
5. Architecture of Data Mining (KDD Process)
6. Tools in Data Mining
7. Difference between Data Mining and Normal Query

Prepared for OAVS Computer Teacher (TGT) preparation and last-minute revision

TOPIC 1

Concept of Data Warehousing

1.1 What is a Data Warehouse?

Definition: A **Data Warehouse** is a large, centralized repository of integrated data collected from multiple sources (databases, operational systems, external files) and organized specifically to support **management decision-making, reporting, and analysis** — rather than day-to-day transaction processing.

Unlike a normal operational database (which handles daily transactions like billing or order entry), a data warehouse stores **historical and summarized data** collected over months or years, used for analysis and business intelligence.

History: *William H. "Bill" Inmon (b. 1945) is an American computer scientist known as the "Father of Data Warehousing." He wrote the first book on data warehousing, held the first conference on the subject, and offered the first classes in it. He coined the standard definition of a data warehouse given above, along with its key characteristics below. His approach is **top-down** — build one central enterprise-wide warehouse first, then derive department-specific data marts from it — often contrasted with Ralph Kimball's **bottom-up** approach of building data marts first. Inmon has authored over 60 books and around 2,000 articles on data warehousing.*

1.2 Key Characteristics (as defined by W. H. Inmon)

Characteristic	Meaning
Subject-Oriented	Organized around major subjects of the business (e.g., "Sales", "Customer", "Product") rather than around applications.
Integrated	Data pulled from many different, inconsistent sources is cleaned and made consistent (same naming, same units, same format).
Time-Variant	Data is stored with a time reference (e.g., daily, monthly, yearly) so trends can be studied over long periods.
Non-Volatile	Once data enters the warehouse, it is not changed or deleted — only read and analyzed. New data is appended.

1.3 Purpose of a Data Warehouse

- To provide a single, unified view of business data for decision-making.
- To support **OLAP** (Online Analytical Processing), reporting, and business intelligence.
- To store historical data for **trend analysis** and forecasting.
- To separate analytical workload from the operational (transactional) system, so daily business is not slowed down.

1.4 Basic Components of a Data Warehouse

Component	Role
Source Data	Operational databases, external files, flat files, legacy systems.
ETL Layer	Extracts, Transforms, and Loads data from sources into the warehouse.

Data Warehouse (Storage)	Central repository storing integrated, historical, subject-oriented data.
Metadata	"Data about data" — describes source, meaning, and structure of stored data.
Data Marts	Smaller, department-specific subsets of the warehouse (e.g., Sales Data Mart).
OLAP / Query & Reporting Tools	Front-end tools used by analysts/managers to query and visualize data.

EXAM POINT

Remember the 4 characteristics using the short form **S-I-T-N** → Subject-oriented, Integrated, Time-variant, Non-volatile. This is a very frequently asked definition question.

REMEMBER

A Data Warehouse is used for **analysis** (read-heavy), while a normal operational database (OLTP) is used for **daily transactions** (read/write-heavy).

COMMON MISTAKE

Students often confuse Data Warehouse with just "a big database." A data warehouse is not simply a large database — it is specifically designed, integrated, and structured for analytical decision-making.

TOPIC 2

Architecture of Data Warehousing

2.1 Three-Tier Architecture

The Data Warehouse Architecture describes how data flows from source systems into the warehouse and then out to the users. It is commonly explained using a **3-Tier Architecture**.

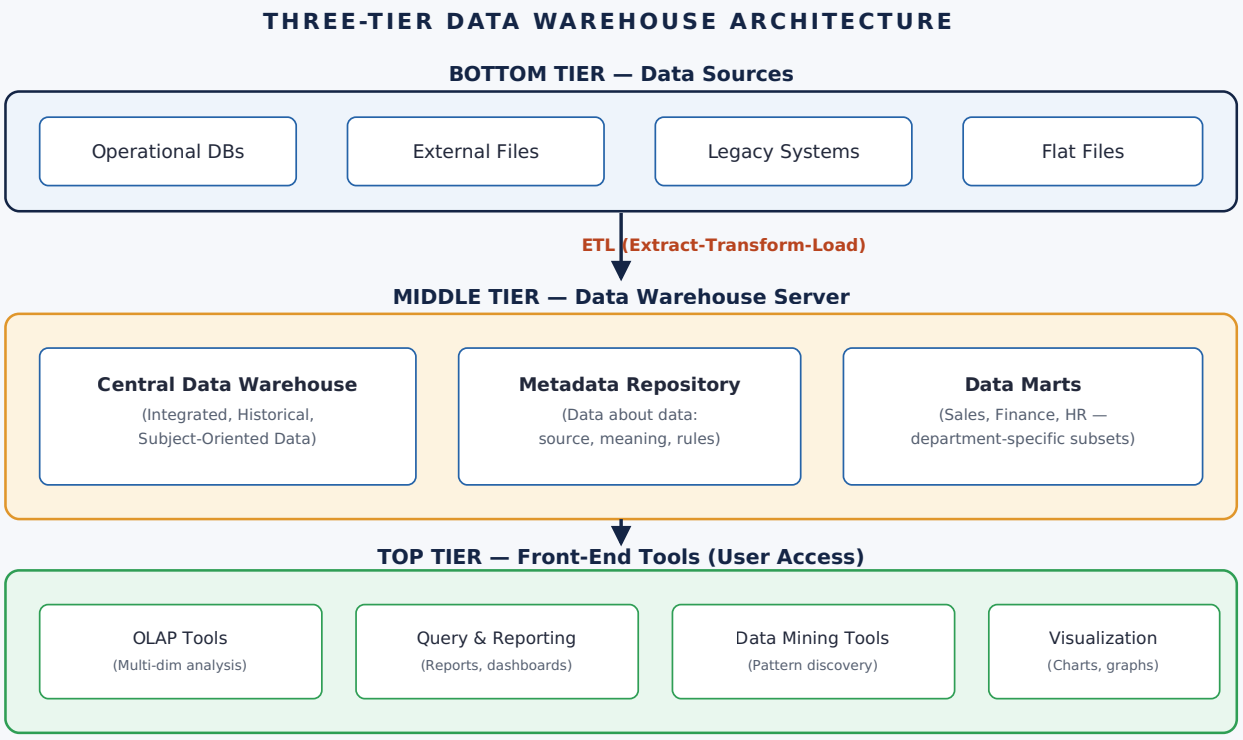


Fig 12.1 — Three-Tier Data Warehouse Architecture showing data flow from sources → warehouse → users

2.2 Explanation of the Three Tiers

Tier	Contains	Function
Bottom Tier	Data sources + ETL	Raw data is extracted from operational databases and external sources, cleaned/transformed, and loaded into the warehouse via ETL tools.
Middle Tier	Data Warehouse Server + Metadata + Data Marts	Stores the integrated, subject-oriented, historical data centrally; an OLAP server may sit here to enable fast multi-dimensional queries.
Top Tier	Front-end tools	Users access data through query tools, reporting tools, OLAP tools, data mining tools, and visualization/ dashboard tools.

EXAM POINT

"Explain the architecture of Data Warehouse with a diagram" is a common long-answer question. Always mention: Source → ETL → Warehouse (+ Metadata + Data Marts) → Front-end tools, in that flow order.

TOPIC 3

Various Tools in Data Warehousing

Data Warehousing tools (used to build/populate/query the warehouse) can broadly be grouped by the stage of the warehousing process they support: ETL tools, OLAP tools, and Reporting/BI tools.

Quick Reference: Key Abbreviations

Before looking at the tools, here is a quick reference to the key abbreviations used throughout this chapter.

Abbreviation	Full Form	Role
OLAP	Online Analytical Processing	Multi-dimensional data analysis (used in Data Warehousing).
OLTP	Online Transaction Processing	Day-to-day operational transactions (used in normal databases).
ETL	Extract, Transform, Load	Moves and cleans data into the warehouse.
BI	Business Intelligence	Reporting/dashboard tools for decision-makers.
KDD	Knowledge Discovery in Databases	Another name for the overall Data Mining process.

EXAM POINT

OLAP tools let you "slice and dice" data — view it from different angles (by region, by time, by product) — which is the core feature examiners often ask you to explain.

3.1 ETL Tools — Extract, Transform, Load

Used to pull data from multiple heterogeneous sources, clean/convert it into a consistent format, and load it into the warehouse.

Tool	Purpose / Role
Informatica PowerCenter	Popular enterprise ETL tool for extracting, transforming, and loading data.
Microsoft SSIS (SQL Server Integration Services)	ETL tool integrated with Microsoft SQL Server for building data pipelines.
Talend Open Studio	Open-source ETL/data-integration tool.
IBM DataStage	Enterprise ETL tool used for large-scale data integration.

3.2 OLAP Tools (Online Analytical Processing)

Allow users to analyze data from multiple dimensions quickly (e.g., sales by region, by product, by time — all at once).

Tool	Purpose / Role
SQL Server Analysis Services (SSAS)	Provides OLAP and data mining functionality on top of SQL Server.
Oracle OLAP	OLAP engine built into Oracle Database for multi-dimensional analysis.

`IBM Cognos`	Business intelligence suite with OLAP analysis capability.

3.3 Reporting & Business Intelligence (BI) Tools

Used by managers/analysts at the front end to view reports, dashboards, and visualizations.

Tool	Purpose / Role
`Tableau`	Data visualization and interactive dashboard tool.
`Microsoft Power BI`	Business analytics tool for interactive reports and dashboards.
`QlikView / Qlik Sense`	Self-service BI and data visualization tool.

REMEMBER

Group tools by function — **ETL** (get data in), **OLAP** (analyze multi-dimensionally), **BI/Reporting** (present to the user). Exams often ask "give any two examples of ETL tools" type questions.

COMMON MISTAKE

Do not confuse Data Warehousing tools (used to build/populate/query the warehouse) with Data Mining tools (used to discover hidden patterns) — they serve different purposes, covered next.

TOPIC 4

Concept of Data Mining

4.1 What is Data Mining?

Definition: Data Mining is the process of discovering hidden, previously unknown, and potentially useful **patterns, trends, and relationships** from large volumes of data — usually data stored in a data warehouse — using statistical, mathematical, and AI-based techniques.

In simple terms, Data Mining goes beyond simply retrieving stored data — it analyzes the data to find patterns that were not directly stored, such as "customers who buy bread also tend to buy butter" (market basket analysis), or "which customers are likely to leave the bank" (churn prediction). Data Mining is also known as **Knowledge Discovery in Databases (KDD)**, since it is essentially the process of extracting useful knowledge from large stores of data.

4.2 Key Characteristics of Data Mining

Characteristic	Meaning
Large Data Volumes	Works on huge datasets, typically pulled from a data warehouse, rather than small, isolated tables.
Pattern & Trend Discovery	Automatically uncovers relationships, patterns, and trends that are not obvious from the raw data.
Predictive in Nature	Goes beyond describing the past — it forecasts future behaviour and outcomes.
Uses Statistical / AI Techniques	Applies methods such as clustering, classification, association rules, and regression.
Non-Trivial Process	Requires multiple steps — cleaning, selecting, and transforming data — before meaningful patterns emerge (see Architecture, next topic).

4.3 Purpose of Data Mining

- To discover hidden patterns and relationships in large datasets.
- To support prediction and forecasting (e.g., future sales, customer churn).
- To help in decision-making, strategy formation, and business planning.
- To classify and group data (e.g., grouping customers by buying behavior).

EXAM POINT

Data Mining is frequently asked as "also known as _____." Remember the synonym: **Knowledge Discovery in Databases (KDD)**.

REMEMBER

Data Mining ≠ simple searching. It **discovers** patterns that were never explicitly stored, and it can **predict** future outcomes — a normal query can do neither (see Topic 7).

TOPIC 5

Architecture of Data Mining

5.1 Data Mining as Part of the KDD Process

Data Mining does not work directly on raw, messy data. It is one step — the core analytical step — inside a larger process called **Knowledge Discovery in Databases (KDD)**. The architecture below shows how data moves from raw storage to useful, actionable knowledge.

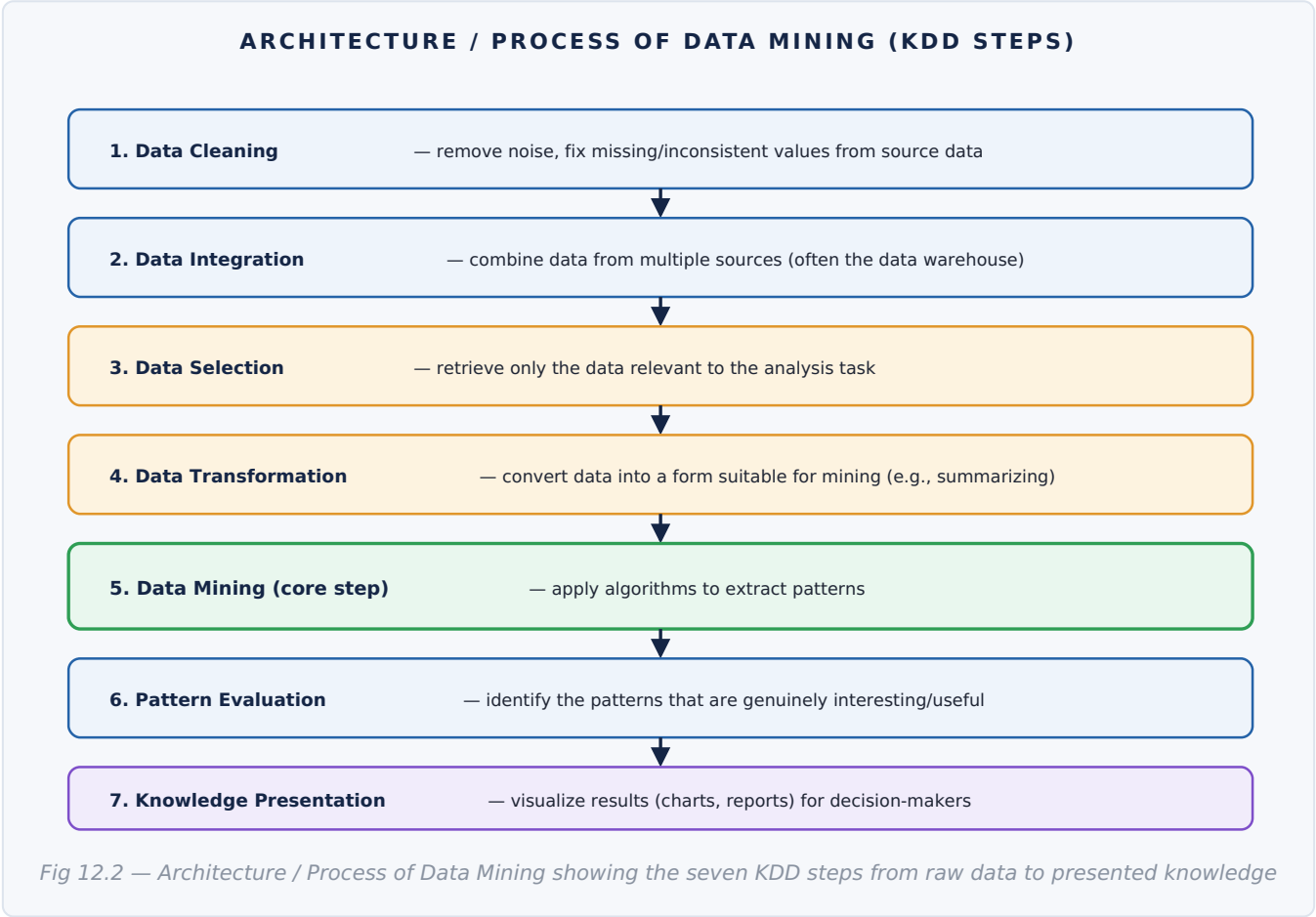


Fig 12.2 — Architecture / Process of Data Mining showing the seven KDD steps from raw data to presented knowledge

5.2 Explanation of the Steps

Step	What Happens
1. Data Cleaning	Removes noise, duplicate records, and fixes missing or inconsistent values.
2. Data Integration	Merges data coming from multiple heterogeneous sources, typically from the data warehouse.
3. Data Selection	Selects only the data relevant to the specific mining task at hand.
4. Data Transformation	Converts/consolidates data into forms appropriate for mining (e.g., aggregation, normalization).
5. Data Mining	The core step — applies algorithms (classification, clustering, association rules, etc.) to extract patterns.
6. Pattern Evaluation	Filters out trivial or irrelevant patterns, keeping only genuinely interesting/useful knowledge.

7. Knowledge Presentation	Presents the discovered knowledge to the user through visualization, reports, or dashboards.

EXAM POINT

Remember all 7 steps in order — "Explain the architecture/process of Data Mining" is a common long-answer question, and marks are often given per correctly-ordered step.

SHORTCUT

Remember the steps as **"C-I-S-T-M-E-P"**: Cleaning → Integration → Selection → Transformation → Mining → Evaluation → Presentation.

TOPIC 6

Tools in Data Mining

Just as Data Warehousing has ETL, OLAP, and BI tools, Data Mining relies on specialized software to carry out the core "Data Mining" step of the KDD process (Topic 5) — applying algorithms like classification, clustering, and association-rule mining without coding them from scratch.

6.1 Major Tools Used in Data Mining

Tool	Purpose / Major Use
WEKA	Open-source tool with a collection of machine learning and data mining algorithms for classification, clustering, and association.
RapidMiner	Visual, drag-and-drop data mining and predictive analytics tool used for data preparation and modeling.
Orange	Open-source tool for data visualization and data mining through visual programming.
KNIME	Open-source platform for data analytics and building data mining workflows visually.
IBM SPSS Modeler	Enterprise-level predictive analytics and data mining tool used widely in business and research.
Oracle Data Mining	Data mining tool built into Oracle Database for pattern discovery directly on stored data.

EXAM POINT

Be ready to name at least 3–4 data mining tools (WEKA, RapidMiner, Orange, KNIME are the most commonly asked) along with one line on their use.

SHORTCUT

Remember tools using "**WRO-K**" — WEKA, RapidMiner, Orange, KNIME — all are popular open-source data mining tools.

TOPIC 7

Data Mining vs Normal Query

7.1 Core Difference

A **normal query** (like an SQL query) simply **retrieves data that already exists** in a database based on conditions specified by the user. It does not discover anything new — it only fetches what is asked for. **Data Mining**, on the other hand, **analyzes** the data to uncover patterns and relationships that are **not explicitly stored** and were previously unknown.

Normal Query Example:
"List all customers who purchased more than ₹10,000 worth of goods in July." — This simply retrieves existing facts.

Data Mining Example:
"Which customers are likely to purchase more than ₹10,000 worth of goods next month?" — This predicts a new, previously unknown fact.

7.2 Detailed Comparison Table

Basis	Normal Query (e.g., SQL)	Data Mining
Purpose	To retrieve specific, already-known data based on given conditions.	To discover hidden, unknown patterns and relationships in data.
Type of Result	Exact answer / list of records matching the condition.	Patterns, trends, clusters, rules, or predictive models.
Data Analysis	Simple retrieval — no deep analysis; works on current data as stored.	Deep statistical/algorithmic analysis of large historical data.
Pattern Discovery	No pattern discovery — user must already know what to ask.	Automatically discovers patterns the user did not explicitly ask about.
Predictive Capability	No prediction — only reports existing facts.	Can predict future trends/behaviour (e.g., forecasting, classification).
User Interaction	User must know exactly what question (query) to ask.	User sets a general goal; the tool explores data and finds **insights** on its own.
Examples	"Show all orders placed in June"; "Find employees with salary > ₹50,000."	"Predict customers likely to churn"; "Find products frequently bought together" (market basket analysis).

***Note:** An **insight** is a meaningful fact or pattern discovered from data analysis that helps in understanding or decision-making.*

7.3 Real-World Illustration

Domain	Normal Query	Data Mining
Retail / Sales	Total sales of a product last month.	Which products are usually bought together (basket analysis).
Banking	List of customers with loan overdue.	Predicting which customers are likely to default on a loan.

Customer Behaviour	Number of customers who visited the store today.	Segmenting customers into groups based on buying habits.

EXAM POINT

A very common question: "Differentiate between Data Mining and a normal database query." Always highlight the core idea — query = retrieval of known facts; mining = discovery of unknown patterns/ predictions.

REMEMBER

Query answers **"What happened?"**; Data Mining answers **"What patterns exist, and what is likely to happen next?"**

COMMON MISTAKE

Do not say "data mining is just a complex query." Data mining uses statistical/AI techniques to find patterns not directly present in the data — a query, however complex, only retrieves stored facts.

Quick Revision Sheet — Data Warehousing & Data Mining

Key Definitions

- **Data Warehouse:** Centralized, integrated, subject-oriented, time-variant, non-volatile repository of data used for analysis and decision-making.
- **Data Mining:** Process of discovering hidden patterns, trends, and relationships from large datasets.
- **ETL:** Extract, Transform, Load — process of moving data from sources into the warehouse.
- **Metadata:** Data about data — describes source, format, and meaning of stored data.
- **Data Mart:** Smaller, department-specific subset of a data warehouse.

Characteristics of Data Warehouse (S-I-T-N)

- Subject-Oriented — organized by business subject
- Integrated — consistent data from many sources
- Time-Variant — stored with historical time reference
- Non-Volatile — data is not updated/deleted, only added

Data Warehouse Architecture Flow (S-E-W-F)

- **S**ource Systems → **E**TL → **W**arehouse (+ Metadata + Data Marts) → **F**ront-end tools (OLAP, Reporting, Mining, Visualization)

Data Mining Architecture / KDD Process (C-I-S-T-M-E-P)

- **C**leaning → **I**ntegration → **S**election → **T**ransformation → **M**ining (core step) → **E**valuation → **P**resentation

Tools at a Glance

Category	Examples
ETL Tools	`Informatica`, `SSIS`, `Talend`, `IBM DataStage`
OLAP Tools	`SSAS`, `Oracle OLAP`, `IBM Cognos`
BI / Reporting Tools	`Tableau`, `Power BI`, `QlikView`
Data Mining Tools	`WEKA`, `RapidMiner`, `Orange`, `KNIME`, `SPSS Modeler`, `Oracle Data Mining`

Data Mining vs Normal Query — One Line Each

- **Normal Query:** Retrieves known, existing data on demand — no pattern discovery, no prediction.
- **Data Mining:** Discovers hidden patterns and predicts future trends using statistical/AI methods.

One-Line Exam Facts

- Data Warehouse supports OLAP; operational databases support OLTP.
- Data in a warehouse is read-only for end users — non-volatile.
- ETL must run before data reaches the warehouse.
- A Data Mart serves a single department; a Data Warehouse serves the whole organization.
- Data Mining is also called "Knowledge Discovery in Databases (KDD)" in some texts.
- Market Basket Analysis is a classic real-world example of data mining.
- A query needs a specific question; data mining finds insights without a specific question.